

Small-signal stability analysis for two-mass and three-mass shaft model of wind turbine integrated to thermal power system[☆]



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ABSTRACT

The integration of large wind power into grid affects the power systems dynamics, operation and control. This paper presents the stability analysis of small-signal models of two-mass and three-mass wind turbine interconnected to thermal power system. The use of two-mass/three-mass model is motivated by the need to deal with flexible modes induced due to intermittent varying of wind speed and low speed shaft stiffness. Initially, eigenvalue analysis is performed to investigate the stability of drive train models and also its union model with thermal system. Next, drive train models of wind turbine and its union model are excited by different wind speed profiles. The decomposition of signal into low frequency oscillation (LFO) modes as collection of intrinsic mode functions (IMFs) computed from Hilbert-Huang Transform (HHT) is achieved. The study suggests excitation of LFO modes in drive train models and the same modes in approximate form can be also found in union model with thermal system.

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1. Introduction

The penetration level of wind turbine is increasing throughout world-wide with total installed capacity of 600 GW globally at the end of 2018, according to preliminary data provided by wind world energy association. It is expected that more than 25% of the world electricity will be generated from wind power by 2030 and save more than 3 billion CO₂ emission annually [1]. This leads to a major contribution in electrical power grid. However, mechanical resonance of wind turbine occurs frequently, as a matter of fact, the torsional vibration of shaft and the communication between mechanical and electrical systems of wind turbine has received wide concern.

Due to large difference in capacity of WTGS (wind turbine generator system) against thermal power plants, the penetration of high wind power may affect the capacity of load flow distribution, resulting into negative damping effect, which leads to the instability of power grid. The wind turbine models have evolved in wide forms by the researchers in order to investigate its integration to the grid [2,3]. As penetration level of wind power generation is expected to increase in future,

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List of abbreviations

LFO	Low frequency oscillation
IMF	Intrinsic mode functions
HHT	Hilbert-Huang Transform
WTGS	Wind turbine generator system
EMD	Empirical mode decomposition
VHM	Vander-Hoven model
HT	Hilbert transform
FFT	Fast Fourier transform
DFIG	Doubly fed induction generator
WT	Wind turbine
MOI	Moment of inertia
ACE	Area control error
UM	Union model
FACTS	Flexible alternating current transmission System
DG	Distributed generation

List of symbols

A	State matrix
B	Control matrix
Γ	Disturbance matrix
$\Delta P_{\text{tie}(1,2)}$	Tie line power between areas
T_0	Tie-line synchronizing coefficient
ACE_i	Area control error
ΔP_{g1}	Governor valve position of area-1
ΔP_{g2}	Governor valve position of area-2
ΔP_{D1}	Changes or deviation in load of area-1
P_{D2}	Changes or deviation in load of area-2
B_i	Frequency bias factor of ith area
M_i	Moment of inertia of generator unit of ith area
D_i	Damping coefficient of generator unit of ith area
Δf_1	Frequency deviation of area-1
Δf_2	Frequency deviation of area-2
ΔP_{t1}	Generator mechanical output of area-1
ΔP_{t2}	Generator mechanical output of area-2
a_{12}	Area size coefficient
M_1	Moment of inertia (MOI) of blades
M_2	MOI of low-speed shaft
M_3	MOI of high-speed shaft.
A_{3U}	State matrix of three mass model
R_i	Regulation or droop characteristic of ith area
P_m	Driving mechanical power
$\Delta \omega_i$	Deviation in angular velocity
$\Delta \theta_i$	Deviation in rotor angle
ΔT_{wt}	Aerodynamic torque (pu)
T_{em}	Electromagnetic torque (pu)
J_r	Turbine rotational inertia,
J_g	Generator rotational inertia,
K_{mec}	Torsional spring
D	Torsional damper
T_{ls}	Low-speed shaft
η_g	Ideal gear box
T_{hs}	High speed shaft
K_i	Stiffness of masses (Nm/rad)
K_{12} and K_{23}	stiffness between masses (Nm/rad)
D_{12} and D_{23}	damping coefficient between two mass (Nms/rad)
D_3	damping coefficient of high-speed shaft.
A_{2U}	State matrix of two mass model
D_1	Damping coefficient of blades

D_2	Damping coefficient of low-speed shaft
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the small-signal stability analysis has been of utmost importance. For analysis of dynamic behavior of WTGS, different types of mass-models like one-mass, two-mass, three-mass and six-mass model have been presented in the past. One-mass model of wind turbine system is not capable to analyze the dynamic behavior of system accurately because it cannot represent the wind dynamics accurately. The modeling of transient behavior of WTGS has been performed using two-mass shaft models [4–6] and three-mass shaft models [7,8]. The authors of [9] have presented nonlinear control of variable speed wind turbine using two mass models. Higher mass modeling of drive train creates a computational problem because of number of stages involved. So, for the sake of simplicity, generally two-mass and three-mass drive train are taken into consideration by researchers for analyzing the transient behavior of WTGS.

The authors of [10] have presented an effect of grid side converter current on dynamics of small signal stability using eigenvalue analysis. Additionally, in study, flexible alternating current transmission system devices (FACTS) have been used to examine its impact on system stability. Further, small signal stability of fractional transmission system having offshore wind is presented in [11]. The authors [12] have presented the small signal dynamics of DFIG based wind turbine system in weak ac grid. Further authors [13] have presented an equivalent modeling of wind farm for small signal stability in weak ac network.

Due to highly dynamic characteristics of wind profile, serious issue related to torsional shaft vibration gets induced, resulting into presence of flexible modes. The modal analysis of a grid connected doubly fed induction generator (DFIG) is presented in [14]. The modal analysis concerned to control mode of DFIG is discussed in [15]. Recently authors [16] have proposed the impact of increased penetration of converter control-based generators on power system modes of oscillation.

In the past, small-signal stability analysis of wind power system was presented in [17,18]. However, impact of using two-mass/three-mass wind turbine model integrated to thermal power system and their induced mode analysis due to wind dynamics as small-signal stability analysis has not been discussed much. The authors of [19] have presented the modal analysis of grid connected DFIG, with focus on the detailed mechanical structural modeling of the wind turbine. The power system dynamics of the generator with the mechanical-electric coupling of the system has been presented in [20]. It is obvious that a large number of wind generators are going to be connected with existing power grid in near future to fulfill the load requirement. Therefore, it is important to analyze the stability of power system, including wind turbine generator system. Further, Authors of [21] have presented a novel coordinated control approach for high penetration of wind energy in smart grid.

In this work, an effort has been made to show the stability issues in small-signal modelling of two-mass and three-mass wind turbine system integrated to thermal-thermal system. The model based on small signal stability analysis is used to study the wind turbine shaft torsional vibration. Due to this reason, this work concentrates on union model of wind turbine system and thermal power system.

In this paper, we propose a new approach to analyze the induced flexible modes in two-mass/three mass wind turbine drive model integrated to thermal power system and its union model-based drive train system. In order to find low frequency oscillation (LFO) signal, Hilbert-Huang Transform (HHT) is applied to analyze the oscillatory modes present in the signals (shaft torque and frequency deviation in area-1 and area-2). Further, two different wind profile have been considered in this study to evaluate the vibrational mode excitation for two-mass /three-mass and its union model. The stability analysis of two-mass, three-mass and its union models is performed using eigenvalue analysis. In order to analyze the relation among the system state variables, participation factor is calculated which gives the relation between modes and corresponding variables. Further, location of poles is also determined for investigation of stability of the system.

The paper is organized as follows. Section II describes the small-signal modelling and stability of thermal power system connected to wind turbine. Section III, describes the modelling of wind profile followed by modal analysis using EMD in section IV. Section V presents the small signal analysis of system. Section VI discusses simulation results on study system. Finally, conclusion is given in section VII.

2. Small-signal modelling and stability of thermal power system connected to wind turbine

The schematic diagram of wind turbine integrated to thermal power system (two-area) is shown in Fig. 1. The output power of WTGS is fed to thermal power system in area-1. The frequency deviation will occur in each area due to mismatch between total generation and load demand. To study the torsional effects in wind turbine model, a small signal stability model is needed.

2.1. Modeling of two-mass shaft wind turbine

The aerodynamic torque is generated in the rotor area, i.e. blade, and this torque is applied through two-mass drive train to generator unit. This generated torque from drive train model is transformed through gears into mechanical torque and is applied to generator shaft. Its output electrical power combines with those of thermal system in area-1. The effect of integration of wind to thermal system will result in excitation of modes in frequency signals, tie-line power and electrical

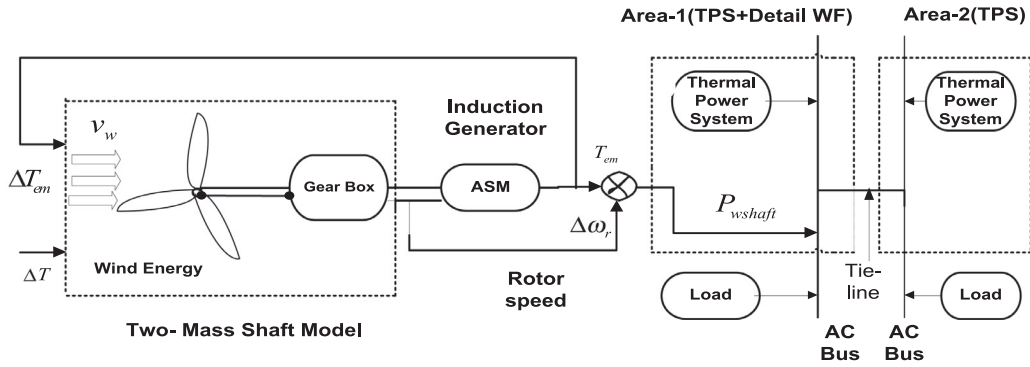


Fig. 1. Configuration of WTGS integrated to thermal power system.

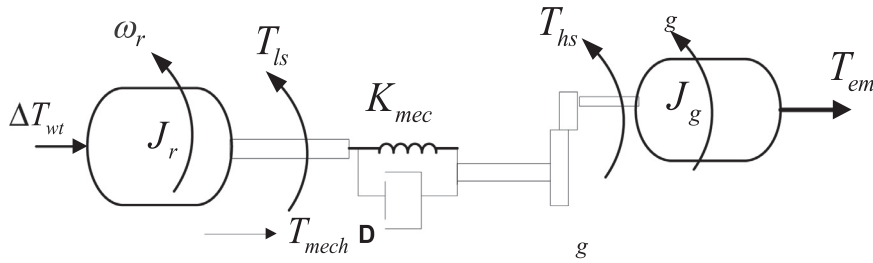


Fig. 2. Schematic of two-mass shaft model for wind turbine.

shaft power. In this section, wind turbine (WT) is modeled as two-mass drive train system as shown in Fig. 2. The details of component of WT drive train model is given as follows.

$$\Delta \dot{\omega}_r = \frac{1}{2J_r} (T_{wt} - T_{mec}) \quad (1)$$

$$\Delta \dot{\omega}_g = \frac{1}{2H_g} (T_{mec} - T_e) \quad (2)$$

$$\Delta \dot{T}_{mec} = D \left(\frac{(T_{wt} - T_{mec})}{2J_r} - \frac{(T_{mec} - T_e)}{2H_g} \right) + K(\omega_r - \omega_g) \quad (3)$$

Eqs. (1)–(3) can be represented as by general Eq. (4)

$$\begin{aligned} \dot{X}_T &= A_T X_T + B_T u_T \\ Y_T &= C_T X_T + D_T u_T \end{aligned} \quad (4)$$

Where, $X_T = [\Delta \omega_r \quad \Delta \omega_g \quad \Delta T_{mec}]^T$

Where ΔT_{wt} aerodynamic torque (pu), T_{em} is electromagnetic torque (pu), J_r turbine rotational inertia, generator rotational inertia J_g , torsional spring K_{mec} and torsional damper D representing low-speed shaft T_{ls} , η_g ideal gear box and high speed shaft T_{hs} . The state-space representation of two-mass shaft model is given by Eqs. (1)–(3).

2.2. Modeling of three-mass shaft wind turbine

A mechanical rotational of wind turbine system possess turbine blades, rotor inertia, low speed shaft, gearbox, high speed shaft and generator rotor inertia. The physical nature of a rotating system can be derived by dividing the system into masses. The accurate results of WT drive train and its dynamic behavior can be found by increasing the number of masses [22]. The transient stability analysis of wind system using different drive train configuration for different applications and have shown that higher masses are more appropriate for transient analysis during fault conditions.

The schematic diagram of WT drive train as a three-mass model is shown in Fig. 3. The presence of gear box in WT drive train model is necessary because the rotor shaft and generator shaft run on different speeds. From Fig. 3, it is clear that gear box inertia is reflected in low speed shaft and high speed shaft. The inertia of high-speed shaft should be added to generator shaft inertia.

The details on symbols are as follows. T_w (pu) represents aerodynamic torque, T_e (pu) represents electromechanical torque, n represents gearbox ratio, M_1 represents moment of inertia (MOI) of blades, M_2 represent MOI of low-speed shaft

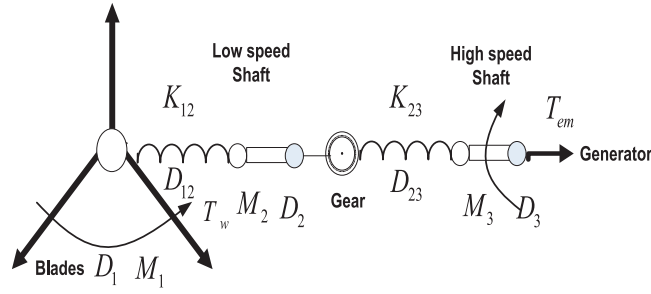


Fig. 3. Schematic of three-mass shaft model for wind turbine.

and M_3 represents the MOI of high-speed shaft. Similarly, D_1 represents the damping coefficient (DC) of blades, D_2 represents the DC of low-speed shaft and D_3 represents the DC of high-speed shaft. K_i is the stiffness of masses (Nm/rad) where $i = 1, 2, 3$, K_{12} and K_{23} are the stiffness between masses (Nm/rad), and D_{12} and D_{23} represent the DC between two mass (Nms/rad)

Further to analyze the small signal stability of three-mass of WT drive train a state space modelling is given in Eqs. (5)–(10), wherein $\Delta\omega_i$ and $\Delta\theta_i$ denotes deviation in angular velocity and rotor angle respectively.

$$\Delta\dot{\theta}_1 = \Delta\omega_1 \quad (5)$$

$$\Delta\dot{\theta}_2 = \Delta\omega_2 \quad (6)$$

$$\Delta\dot{\theta}_3 = \Delta\omega_3 \quad (7)$$

$$M_1 \Delta\dot{\omega}_1 = \Delta T_w - D_1 \Delta\omega_1 - D_{12}(\Delta\omega_1 - \Delta\omega_2) - K_{12}(\Delta\theta_1 - \Delta\theta_2) \quad (8)$$

$$M_2 \Delta\dot{\omega}_2 = -D_2 \Delta\omega_2 - D_{12}(\Delta\omega_2 - \Delta\omega_1) - D_{23}(\Delta\omega_2 - \Delta\omega_3) - K_{12}(\Delta\theta_1 - \Delta\theta_2) - K_{23}(\Delta\theta_2 - \Delta\theta_3) \quad (9)$$

$$M_3 \Delta\dot{\omega}_3 = \Delta T_e - D_3 \Delta\omega_3 - D_{23}(\Delta\omega_3 - \Delta\omega_2) - K_{23}(\Delta\theta_3 - \Delta\theta_2) \quad (10)$$

Eqs. (5)–(10) can be represented in standard state-space form in Eq. (11):

$$\begin{aligned} \dot{X}_{TM} &= A_{TM} X_{TM} + B_{TM} u_{TM} \\ Y_{TM} &= C_{TM} X_{TM} + D_{TM} u_{TM} \end{aligned} \quad (11)$$

Where, $X_{TM} = [\Delta\theta_1 \quad \Delta\theta_2 \quad \Delta\theta_3 \quad \Delta\omega_1 \quad \Delta\omega_2 \quad \Delta\omega_3]^T$

2.3. Modeling of thermal -Thermal power system

The mathematical representation of two-area system is given as:

$$\Delta\dot{f}_1 = -\frac{1}{T_{p1}} \Delta f_1 + \frac{K_{p1}}{T_{p1}} \Delta P_{t1} - \frac{K_{p1}}{T_{p1}} \Delta P_{tie(1,2)} - \frac{K_{p1}}{T_{p1}} \Delta P_{D1} \quad (12)$$

$$\Delta\dot{P}_{t1} = -\frac{1}{T_{t1}} \Delta P_{t1} + \frac{1}{T_{t1}} \Delta P_{g1} \quad (13)$$

$$\Delta\dot{P}_{g1} = -\frac{1}{R_1 T_{g1}} \Delta f_1 - \frac{1}{T_{g1}} \Delta P_{g1} + \frac{1}{T_{g1}} u_1 \quad (14)$$

$$\Delta\dot{f}_2 = -\frac{1}{T_{p2}} \Delta f_2 + \frac{K_{p2}}{T_{p2}} \Delta P_{t2} - \frac{K_{p2}}{T_{p2}} \Delta P_{tie(1,2)} - \frac{K_{p2}}{T_{p2}} \Delta P_{D2} \quad (15)$$

$$\Delta\dot{P}_{t2} = -\frac{1}{T_{t2}} \Delta P_{t2} + \frac{1}{T_{t2}} \Delta P_{g2} \quad (16)$$

$$\Delta\dot{P}_{g2} = -\frac{1}{R_2 T_{g2}} \Delta f_2 - \frac{1}{T_{g2}} \Delta P_{g2} + \frac{1}{T_{g2}} u_2 \quad (17)$$

$$\Delta \dot{P}_{tie} = 2\pi T^0 \Delta f_1 - 2\pi T^0 \Delta f_2 \quad (18)$$

$$ACE_1 = B_1 \Delta f_1 + \Delta P_{tie(1,2)} \quad (19)$$

$$ACE_2 = B_2 \Delta f_2 + \Delta P_{tie(1,2)} \quad (20)$$

Where, $\frac{1}{K_{pi}} = D_i$ & $\frac{T_{pi}}{K_{pi}} = M_i$, subscript i refers to area 1 & 2. ACE_i , the area control error (ACE), $\Delta P_{tie(1,2)}$ the tie-line power between areas and T_0 is tie-line synchronizing coefficient between two areas.

The states, Δf_1 and Δf_2 is frequency deviation, ΔP_{t1} and ΔP_{t2} is mechanical output, ΔP_{g1} and ΔP_{g2} is valve position, ΔP_{D1} and ΔP_{D2} are load perturbation in respective areas. The system parameter nomenclature are as follows; B_i, D_i, T_{ti}, T_{gi} & R_i are the frequency bias coefficient of i th area, damping coefficient of generator unit, time constant of governor, time constant of turbine and speed droop respectively. The system parameter of thermal power system is given in appendix section.

From Eqs. (12)–(20), the complete state space model can be represented as:

$$\begin{aligned} \dot{x}_{Th} &= A_{Th} x_{Th} + B_{Th} u_{Th} \\ Y_{TH} &= C_{TH} x_{TH} + D_{TH} u_{TH} \end{aligned} \quad (21)$$

$$\text{Where, } x_{TH} = [\Delta f_1 \quad \Delta P_{t1} \quad \Delta P_{g1} \quad \Delta f_2 \quad \Delta P_{t2} \quad \Delta P_{g2} \quad \Delta P_{tie} \quad ACE_1 \quad ACE_2]^T$$

2.4. Union model of system

To investigate the induced modes in the mechanical shaft vibration of wind turbine system due to wind speed, it is essential to develop a complete union small-signal stability model of the system; two-mass and three-mass shaft models integrated to thermal power system.

By solving Eqs. (1)–(3) and Eqs. (12)–(20), union model of two-mass WT connected to thermal power system is given as:

$$\dot{X} = A_{2U} X + B u \quad (22)$$

where, state variables $X = [X_T \quad X_{TH}]^T$, A_{2U} is the state matrix of union model (UM) of two-mass WT integrated to thermal power system.

Next, solving Eqs. (5)–(10) and Eqs. (12)–(20), the UM of three-mass wind turbine integrated to thermal power system is obtained as:

$$\dot{X}_h = A_{3U} X_h + B u \quad (23)$$

where, state variables $X_h = [X_{TM} \quad X_{TH}]^T$ and A_{3U} is the state matrix of UM of three-mass WT integrated to thermal power system.

2.5. Small signal stability model

A union model of wind turbine generator system integrated to thermal power system is obtained to solve the equations of drive train model, generator model and thermal power system in single entity. The coupling of mathematical equations between the components to derive the complete small signal model is shown in Fig. 4. The complete state-space representation is given as:

$$\dot{X}_s = X_s + B u \quad (24a)$$

$$\text{Where, } X_s = [X_T \quad X_{TM} \quad X_{TH} \quad X_G]^T$$

3. Modeling of wind profile

The intermittent property of wind profile is modeled in this section. Vander-Hoven [23] is used in this study to model the wide band variation in wind speed. The spectral power of wind speed is computed in range from 0.0007 to 900 cycle/hours. The numerical wind speed procedure based on sampling the spectrum of wind is developed in [23]. Consider $\omega_0, i = 1$ to $(N + 1)$ is discrete angular frequency and $P_{vv}(\omega_i)$ the corresponding power spectral density.

The harmonic frequency amplitude is represented as

$$M_i = \frac{2}{\pi} \sqrt{\frac{1}{2} [(P_{vv}(\omega_i) + P_{vv}(\omega_{i+1}))] (\omega_{i+1} - \omega_i)} \quad (24b)$$

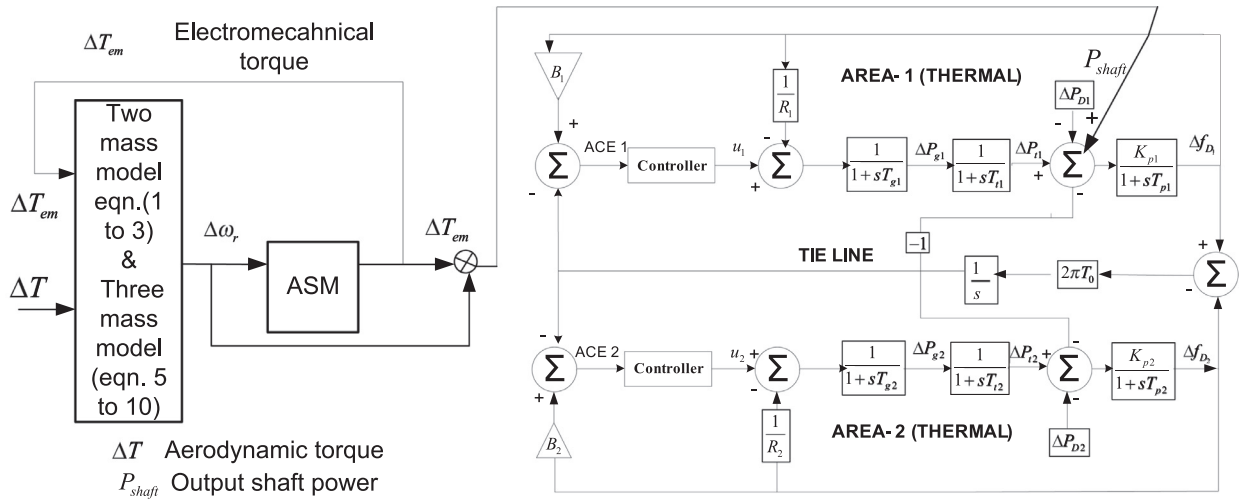


Fig. 4. Interactions between the components.

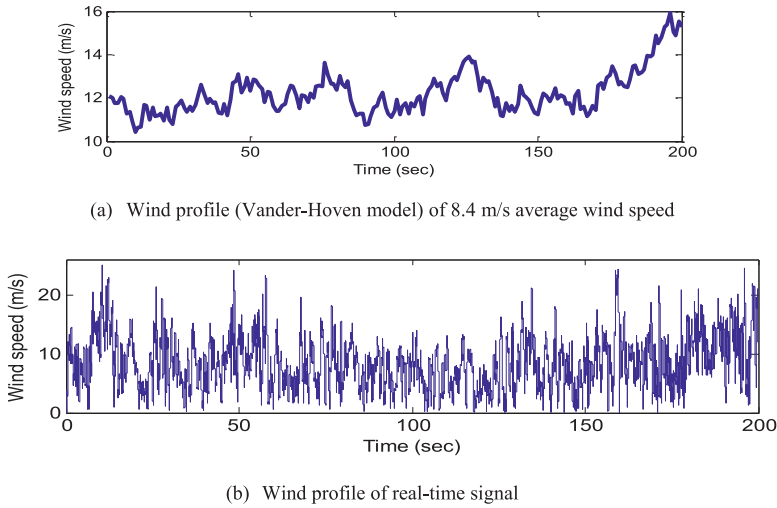


Fig. 5. Two different wind profiles.

And phase φ_0 is randomly obtained, with an even distribution in the domain $[\pi, -\pi]$. The wind speed is computed as per the criteria

$$v_{wind}(t) = \sum_{i=0}^N M_i \cos(\omega_i t + \phi_i) \quad (25)$$

With $N=55$ for the sampling frequency of the Vander-Hoven characteristic, the following discrete frequencies are considered as $f_i = i \times 10^4$ Cycle/hour for $i = 1, 2, \dots, 9$ and $j = -3, -2, -1, 0, 1, 2$.

Fig. 5(a) shows the wind profile modeled by Vander-Hoven model (VHM) considering wide variation of wind speed with average wind speed of 8.4 m/s. Further, wind profile for real time signal [24] is shown in Fig. 5(b).

4. Mode analysis technique using EMD

This section describes the steps adopted to analyze the LFO signals using EMD technique.

Step 1: In the first step, the signal is decomposed into sets of intrinsic mode function [IMFs] using EMD method.

Step 2: Compute the instantaneous frequency and amplitude of each IMF using HHT algorithm. After computation of IMFs via EMD, apply Hilbert transform [HT] to the interested IMF. The oscillation mode for the physical characteristics of signal can be represented in complex form as:

$$z(t) = x(t) + iy(t) = A(t)e^{i\phi(t)} \quad (26)$$

Table 1
Eigen value of Matrix A_T : Two mass Model.

No	Eigen values	Oscillation frequency (Hz)	Damping
$\lambda_{1,2}$	$-0.0746 \pm 1.870i$	0.30	0.00379
λ_3	0	0	0

Table 2
Eigen value of Matrix A_{2U} : union model (UM) of two -mass wind turbine connected to thermal power system.

No	Eigen values	Oscillation frequency (Hz)	Damping
$\lambda_{1,2}$	0	0	0
$\lambda_{3,4}$	$-0.0746 \pm 1.8701i$	0.30	0.00379
λ_5	-0.0029	0.00045	1
λ_6	-13.0675	2.08	1
λ_7	-13.0517	2.08	1
$\lambda_{8,9}$	$-0.3801 \pm 3.1880i$	0.511	0.118
$\lambda_{10,11}$	$-0.9913 \pm 2.2607i$	0.40	0.0402
λ_{12}	0	0	0

The above signal is the representation of time dependent phasor or in complex plane with instantaneous amplitude and instantaneous phase given as:

$$A(t) = \sqrt{x(t)^2 + y(t)^2}, \phi(t) = \tan^{-1} \frac{y(t)}{x(t)}$$

$$f(t) = \frac{\left[\frac{d\phi}{dt} \right]}{2\pi} \quad (27)$$

4.1. Hilbert Transform (HT)

Hilbert transform is actually the relationship between the real and imaginary parts of fast fourier transform [FFT] of one-sided function. Hilbert transform $H[s(t)]$ of a signal $s(t)$ is given by:

$$H[s(t)] = \int_{-\infty}^{\infty} \frac{s(\tau)}{s(t - \tau)} d\tau \quad (28)$$

where, t is time and τ is translation parameter.

Hilbert transform is a frequency independent time domain involution that maps real time-domain value into another value. It is also called 90° phase shifter and does not affect non-stationary characteristic of a modulating signal. Actually, any mode excitation results in modulation; therefore, demodulation is required for obtaining signature related to excited mode of oscillation. This can be obtained analytically by following equation:

$$B(t) = s(t) + iH[s(t)] = be^{i\phi(t)} \quad (29)$$

$b(t) = \sqrt{s^2(t) + H^2[s(t)]}$, $\phi(t) = \arctan \frac{H[s(t)]}{s(t)}$ and $i = \sqrt{-1}$, $b(t)$ is the envelope of $B(t)$.

$\hat{B}(i\omega)$ the Fourier transform of signal $B(t)$ and its properties are given by:

$$\hat{B}(i\omega) = \begin{cases} 2\hat{s}(i\omega) & 0 \leq \omega \\ 0 & \omega < 0 \end{cases} \quad (30)$$

where, $\hat{s}(i\omega)$ is Fourier transform of $s(t)$ and ω is angular frequency of $\hat{B}(i\omega)$.

5. Small-signal analysis

To study the dynamic behavior of drive train model and its union model integrated to thermal power system, eigenvalue analysis and participation factor is computed to know the relation between modes and state variable. In this section, eigenvalue analysis of two-mass/three-mass and its union model is presented.

5.1. Eigenvalue analysis of two-mass and its union model

The eigenvalue analysis, oscillation frequencies and damping of matrix A_T for two-mass model and its union model matrix A_{2U} is given in Tables 1 and 2 respectively. From Table 1, it is clear that there is only one oscillation mode present in two-mass model, whereas, from Table 2 in union model, three oscillation modes are present. From Tables 1 and 2 it can be concluded that system is stable since all eigenvalues have negative real parts. In Table 1 only one oscillation mode is

Table 3

Participation factors of state variables for two mass model connected to Thermal power system.*

	$\lambda_{1,2}$	$\lambda_{3,4}$	λ_5	λ_6	λ_7	$\lambda_{8,9}$	$\lambda_{10,11}$	λ_{12}
$\Delta\omega_r$	0	0.3569	0.2866	0	0	0	0	0
$\Delta\omega_g$	0	0.1435	0.7134	0	0	0	0	0
ΔT_{mec}	0	0.5003	0	0	0	0	0	0
Δf_1	0	0	0	0.0209	0.0178	0.2804	0.2866	0.0682
ΔP_{t1}	0	0	0	0.0257	0.0227	0.1193	0.2584	0.2971
ΔP_{g1}	0	0	0	0.4792	0.4342	0.0364	0.0599	0.0333
Δf_2	0	0	0	0.0189	0.0198	0.2796	0.2873	0.0685
ΔP_{t2}	0	0	0	0.0232	0.0252	0.1187	0.2586	0.2978
ΔP_{g2}	0	0	0	0.4321	0.4814	0.0363	0.0599	0.0334
ΔP_{tie}	0	0	0	0	0.0012	0.2896	0	0.4751
ACE_1	0.9	0	0	0	0	0	0	0
ACE_2	0	0.9	0	0	0	0	0	0

* Bold specifies that eigenvalues are highly sensitive to the corresponding state variables.

Table 4Eigen value of Matrix A_{TM} : Three mass model.

No	Eigen values	Oscillation frequency (Hz)	Damping
$\lambda_{1,2}$	$-0.8077 \pm 1.1865i$	0.23	0.7
λ_3	0	0	0
$\lambda_{4,5}$	$-0.3691 \pm 0.249273i$	0.09	0.829
λ_6	-0.5203	0.08	1

Table 5Eigen value of Matrix A_{3U} : union model (UM) of three -mass wind turbine connected to thermal power system.

No	Eigen values	Oscillation frequency (Hz)	Damping
λ_1	-0.08	0.127	1
λ_2	-0.05	0.079	1
$\lambda_{3,4}$	$-0.406 \pm 1.50i$	0.246	0.258
$\lambda_{5,6}$	$-0.22 \pm 0.501i$	0.087	0.406
λ_7	0	0	0
λ_8	-0.477	0.075	1
$\lambda_{9,10}$	$-13.0567 \pm 0.0618i$	2.085	1
$\lambda_{11,12}$	$-0.3814 \pm 3.4964i$	0.560	0.108
$\lambda_{13,14}$	$-1.2523 \pm 1.7425i$	0.3423	0.584
λ_{15}	-0.719	0.1144	1

present whereas in Table 2, two oscillation modes are determined. The mode corresponding to eigenvalue $\lambda_{3,4}$ is poorly damped. The computed participation factor that discloses the strong relation between mode and state variables is given in Table 3. The numerical values in bold faced in said table indicates highly sensitivity of eigenvalues to the corresponding variable. The eigenvalues $\lambda_{1,2}$ is highly sensitive to area control error, i.e. ACE_1 and ACE_2 with participation factor (0.9), while $\lambda_{3,4}$ is highly sensitive to ΔT_{mec} and $\Delta\omega_r$. These are mainly affected by shaft torque and rotor speed with involvement of participation factor of 0.3569 & 0.5003 respectively. Further, λ_5 is highly sensitive to $\Delta\omega_g$, so it is directly influenced by generator speed with participation factor of 0.7134. Similarly, λ_6, λ_7 remain sensitive to gate valve position of both area P_{g1} and P_{g2} . In addition, $\lambda_{8,9}$ is sensitive to mainly frequency deviation and tie line power deviation i.e. $\Delta f_1, \Delta f_2$ and ΔP_{tie} with participation factor of 0.2804, 0.2796 & 0.2896 respectively. The eigenvalue $\lambda_{10,11}$ is highly sensitive to Δf_1 and Δf_2 having oscillation frequency of 0.40Hz with participation factor of 0.2873 & 0.2866 respectively. Lastly, λ_{12} is highly sensitive to turbine mechanical power $\Delta P_{t1}, \Delta P_{t2}$ and ΔP_{tie} .

The natural frequencies of multi drive train WTG system are equal to the masses minus one. Therefore, natural frequency of WT system for two-mass model consists of one oscillation mode of 0.30Hz and its union model with thermal power system.

5.2. Eigen value analysis of three-mass and its union model

The eigenvalue analysis, oscillation frequencies and damping of system matrix A_{TM} for three-mass model and its union model matrix A_{3U} are given in Tables 4 and 5 respectively. From these tables, it can be concluded that the system is stable since all eigenvalues have negative real parts. From Table 4, it is clear that two oscillation modes are present in three-mass drive train model. On other hand, in union model of three-mass WT, five oscillation modes are present (Table 5). The participation factor calculated for union model is given in Table 6. It can be found that $\lambda_{1,2}$ is sensitive to ACE_1, ACE_2 .

Table 6

Participation factor of state variables for three mass model connected to thermal power system.*

	λ_1	λ_2	$\lambda_{3,4}$	$\lambda_{5,6}$	λ_7	λ_8	$\lambda_{9,10}$	$\lambda_{11,12}$	$\lambda_{13,14}$	λ_{15}
$\Delta\theta_1$	0	0	0.0431	0.3226	0.2545	0.1101	0	0	0	0
$\Delta\theta_2$	0	0	0.3937	0.0248	0.3818	0.1143	0	0	0	0
$\Delta\theta_3$	0	0	0.1228	0.2382	0.3636	0.0111	0	0	0	0
$\Delta\omega_1$	0	0	0.0437	0.3351	0	0.2894	0	0	0	0
$\Delta\omega_2$	0	0	0.3876	0.0121	0	0.3082	0	0	0	0
$\Delta\omega_3$	0	0	0.1225	0.2174	0	0.4177	0	0	0	0
Δf_1	0	0	0	0	0	0	0.0194	0.2716	0.2954	0.0333
ΔP_{t1}	0	0	0	0	0	0	0.0243	0.1007	0.4414	0.0697
ΔP_{g1}	0	0	0	0	0	0	0.4580	0.0326	0.0831	0.0105
Δf_2	0	0	0	0	0	0	0.0194	0.2719	0.2951	0.0321
ΔP_{t2}	0	0	0	0	0	0	0.0243	0.1574	0.2371	0.3106
ΔP_{g2}	0	0	0	0	0	0	0.4583	0.0510	0.0447	0.0469
ΔP_{tie}	0	0	0	0	0	0	0.0024	0.2495	0.0822	0.6573
ACE_1	.98	0	0	0	0	0	0	0	0	0
ACE_2	0	.98	0	0	0	0	0	0	0	0

* Bold specifies that eigenvalues are highly sensitive to the corresponding state variables.

Further, $\lambda_{3,4}$ is highly sensitive to $\Delta\theta_2$ and $\Delta\omega_2$, so it is mainly affected by rotation angle deviation and angular velocity deviation. In addition, $\lambda_{3,4}$ is also associated with $\Delta\theta_3$ and $\Delta\omega_3$ to high speed shaft up with participation factor of 0.1128 and 0.1225 respectively. $\lambda_{5,6}$ being highly sensitive to $\Delta\omega_1$ & $\Delta\theta_1$, is directly affected by turbine angular velocity and rotor angle deviation. Similarly, λ_7 is highly sensitive to state variables $\Delta\theta_3$ and $\Delta\theta_2$, so it is mainly affected by the torsion angle and rotor angle deviation. λ_8 is highly sensitive to state variables $\Delta\omega_2$ & $\Delta\omega_3$, being affected by speed of the low-speed shaft and high-speed shaft to an extent of participation factor of 0.3082 and 0.4177 respectively. In addition, $\lambda_{9,10}$ is highly sensitive to gate valve position ΔP_{g1} , ΔP_{g2} . $\lambda_{11,12}$ is highly sensitive to Δf_1 , Δf_2 & ΔP_{tie} with participation factor of 0.2716, 0.2719 & 0.2415 respectively. Further, $\lambda_{13,14}$ is sensitive to Δf_1 , Δf_2 & ΔP_{t1} . λ_{15} is highly sensitive to ΔP_{t2} & ΔP_{tie} . Thus from above discussion, natural frequency of wind turbine system consists of two oscillation modes of 0.23 & 0.09 Hz. Its union model with thermal power system also consists of same range of oscillation mode (0.246 & 0.087 Hz) corresponding to turbine angular velocity and rotor angle deviation state variables.

6. Simulation results and discussion

In this section, dynamic simulation results of drive train WT model connected to thermal-thermal system with wind variation is discussed. The flexible modes induced in generator shaft and frequency deviation of each area are analyzed. The details are given in following subsections.

6.1. Mode excitation in two-mass model

Due to variation in wind speed, torsional vibration can be observed in generator shaft. In order to determine the induced modes in signals, HHT technique is applied to decompose signal into intrinsic mode function called IMF and thereby compute the instantaneous frequency. The generated torque and its frequency spectrum for two-mass model on excitation by wind speed profile are shown in Fig. 6. The FFT used to compute the frequency spectrum of generated torque signal, indicates oscillation modes of 0.80 Hz, 0.43 Hz & 0.32 Hz.

Further, Fig. 7 shows the decomposition of generator shaft signal into IMF using HHT technique. The mean value of interest IMF level (corresponding to low frequency mode of oscillation) indicates instantaneous frequency of 0.322, 0.448, 0.82, 0.79, 0.88 & 1.25 Hz in Fig. 8. Out of these, some of the modes (0.322, 0.448 and 0.79 Hz) are also suggested on computation via FFT (Fig. 6b)). This confirms the presence of oscillation frequency (0.3 Hz) of two mass and its union models with those obtained in the time domain signal.

6.2. Mode excitation in three-mass model

In this section, the mode excitation due to wind speed profile in three-mass model is analyzed. The excited modes are estimated via performing FFT analysis (frequency spectrum) and decomposition of IMF on generator shaft signal. This is shown in Fig. 9. The FFT indicates oscillation frequency of 0.59, 0.24 Hz and 0.3440 Hz. Fig. 10 shows the decomposition of generator shaft signal into IMF using HHT method. The mean value of interest IMF level suggests instantaneous frequency of 0.34, 0.599, 0.98, 0.76, 0.853 & 1.086 Hz as given in Fig. 11. This confirms the oscillation mode (0.34 & 0.59 Hz) with those discussed above (Fig. 9(b)) for three-mass model. From discussion given in above section, natural frequencies of three-mass model consists of two oscillation modes (0.23 Hz & 0.09 Hz) and same have been found in union model as well.

So, from the studies carried out on two-mass and three-mass models, one can conclude that lower range of oscillation modes do exist in their systems.

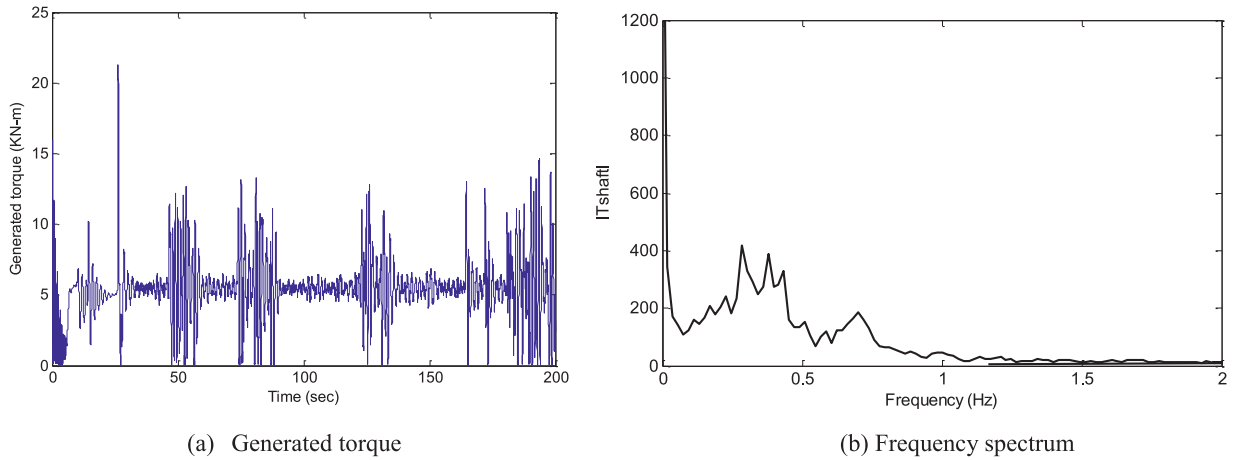


Fig. 6. Generated torque and its frequency spectrum for two mass model.

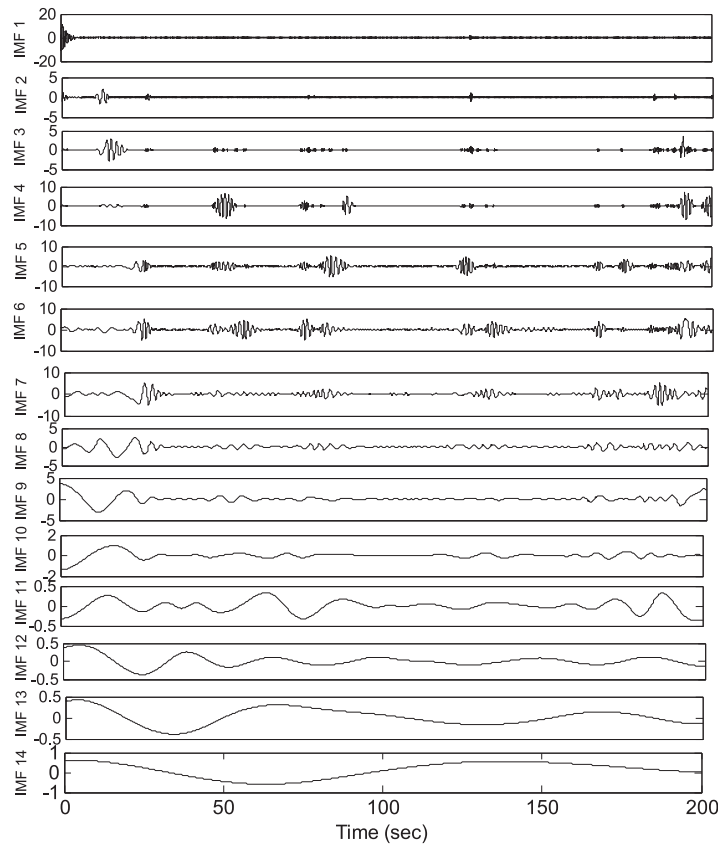


Fig. 7. Decomposed IMF for two- mass model.

6.3. Two –mass model-based WT connected to thermal power system

The penetration of wind power in grid is expected to increase in future. The system having interconnection of wind turbine to thermal power system [25], some of the low frequency modes are likely to be observed on electrical side as well. So, in this section, we discuss the mode excitation for two-mass model WT connected to thermal power system.

Case (a): 0.01 pu step change in load-

The frequency deviation for area-1 & area-2, caused due to step load disturbance is shown in Figs. 12 and 13 respectively. The step load disturbance of 0.01 pu is applied in area-1. With varying wind and load perturbation, the frequency of each

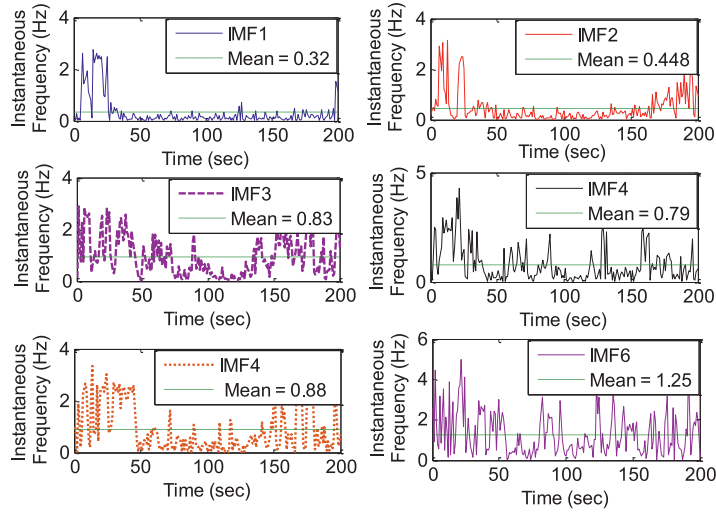


Fig. 8. Instantaneous frequency of interest IMF for two-mass model.

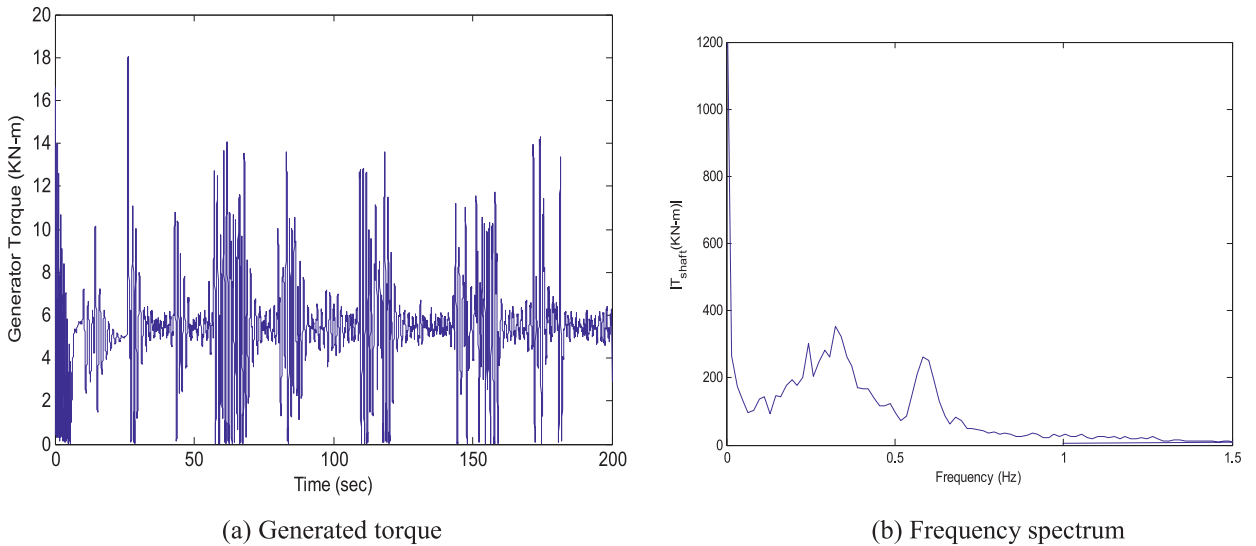


Fig. 9. Generated torque and its frequency spectrum for three -mass model.

area varies according to variation in wind profile. On other hand, in thermal system (area-2) frequency deviation remains minimum. The controller gain of area-2 is optimized using PSO for PI controller. The FFT spectrum of frequency signal in area-1 & area-2 is shown in Fig. 14.

The estimated instantaneous frequency of interest IMF is illustrated in Fig. 15. Mean value of instantaneous frequency of interest IMF level for area-1 is 0.2852, 0.8109, 0.3418, 0.88, 0.54 & 1.06 Hz, while for area-2, it is 0.30, 0.59, 0.94, 0.76, 1.08, and 1.36 Hz. It is indicated that approximately 0.2852, 0.34 & 0.51 Hz are observed in area-1 frequency when computed using FFT (Fig. 14(a)) and IMF. Similarly, excitation of approximately 0.304 Hz and 0.59 Hz can be found in area-2, when computed using FFT and IMF. So, from over all study on two-mass based WT connected to thermal power system, it is clear that low oscillation modes exist in generated shaft power and the same is also reflected on electrical side.

The area control error for area-1 and area-2 is given in Fig. 16. Further, in order to check the overall stability of different drive train models, location of poles for two mass/three mass model and its union model is depicted in Fig. 17. All the poles are located on left half s-plane, confirming the system stability.

6.4. Mode excitation analysis with real-time wind profile

In this section real-time wind profile is considered for mode excitation in two-mass and three-mass models. Due to wind gust presence, flexible modes will get induced on wind turbine and same may be observable on thermal power system side

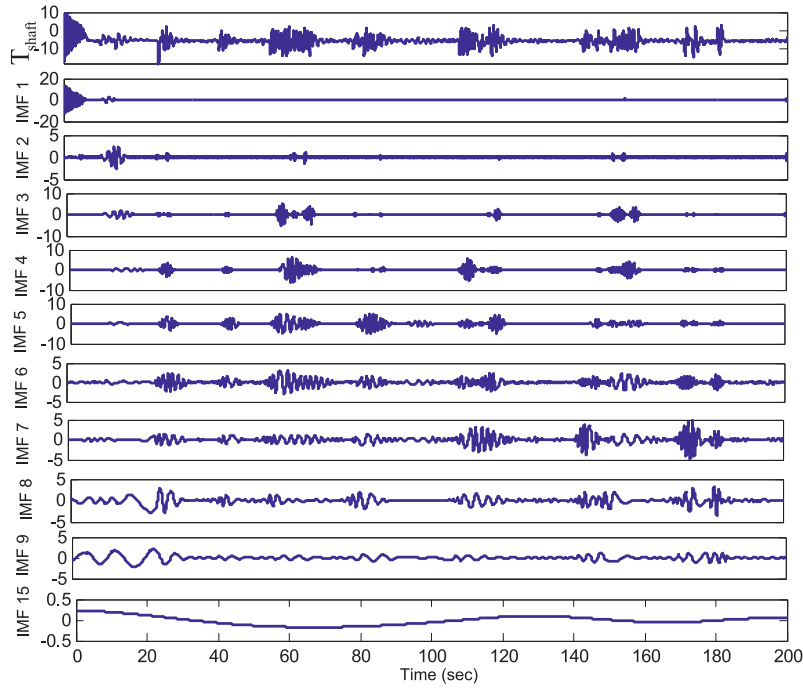


Fig. 10. Decomposed IMF for three mass model.

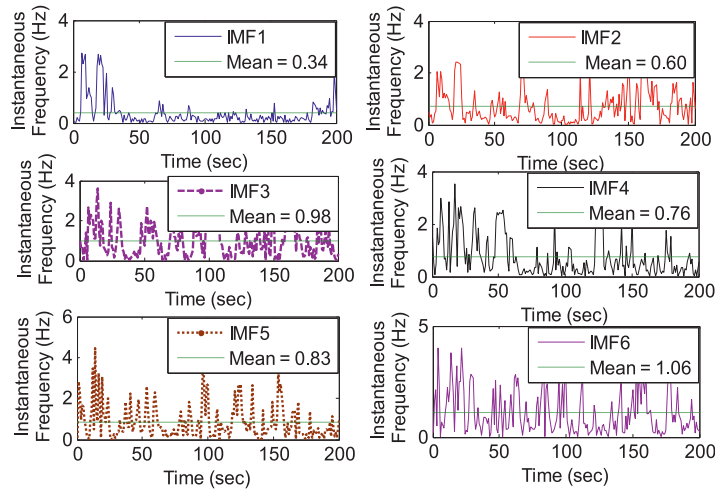


Fig. 11. Instantaneous frequency of interest IMF for three-mass model.

too. Table 7 presents the excited modes due to different wind profile. The italic bold indicates the approximately same modes present in both mechanical and as well as electrical sides.

6.5. Additional discussion

In this study an effort has been made to discuss the presence of torsional vibrational characteristics of wind turbine system on a thermal power system. Firstly, a small signal stability model is developed for different drive train models, and its union model when integrated to thermal power system. The stability is checked via eigenvalue analysis. Further, to understand the exact relation between modes and corresponding state variable, participation factor is computed. The excitation of flexible modes due presence of wind gust is achieved via varying nature of wind profiles. It is found that oscillation mode present in drive train model is very close to those excited in union model having integration with thermal power system. This is verified from estimation of instantaneous frequency on IMFs. As a matter of fact, with increase in number of mass in the modeling of wind system, the number of natural oscillation frequency (mode) also increases. It

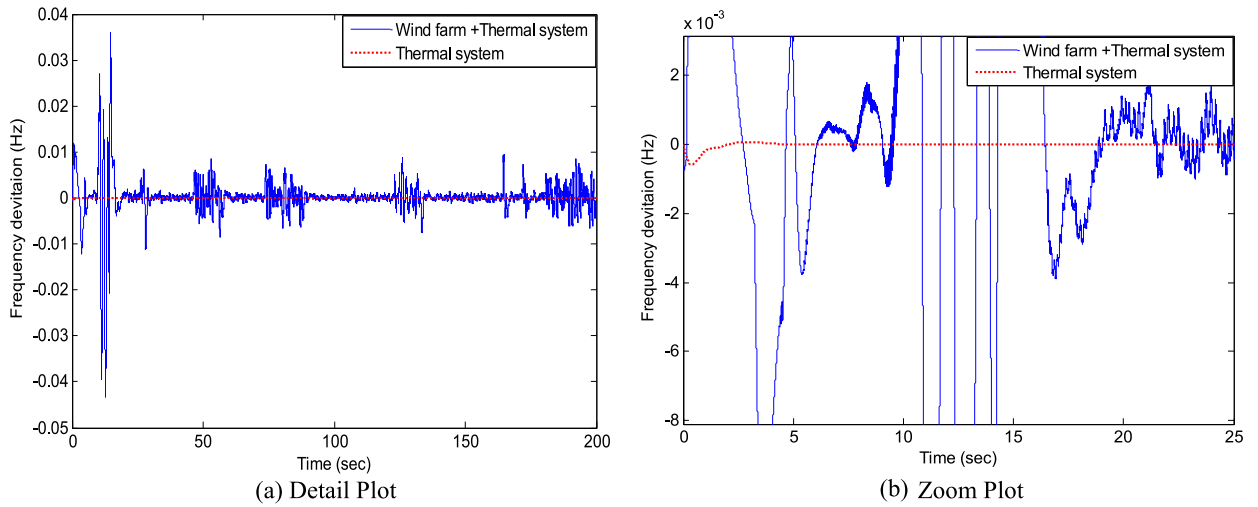


Fig. 12. Frequency deviation area-1.

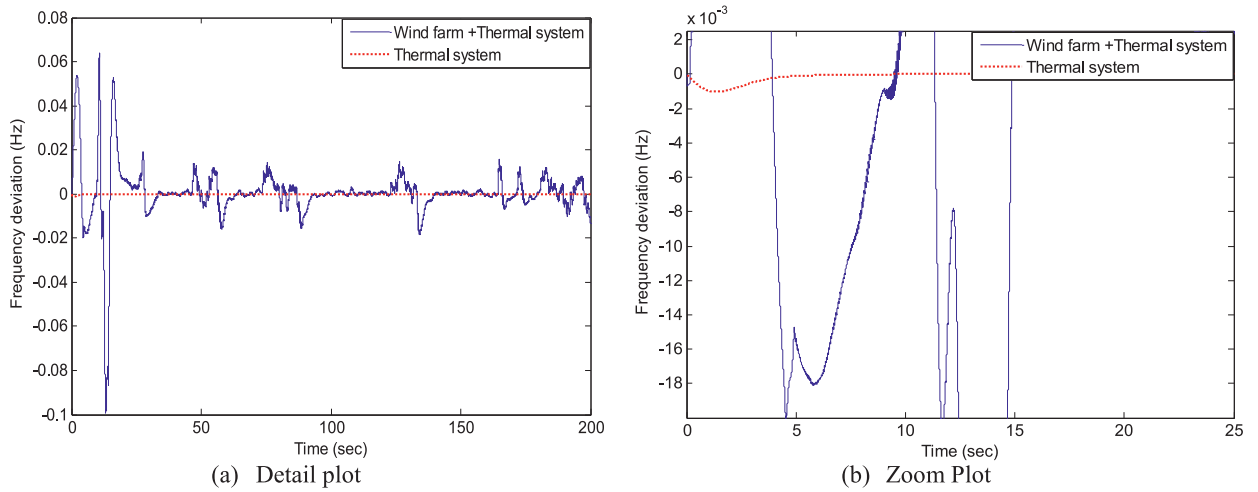


Fig. 13. Frequency deviation of area-2.

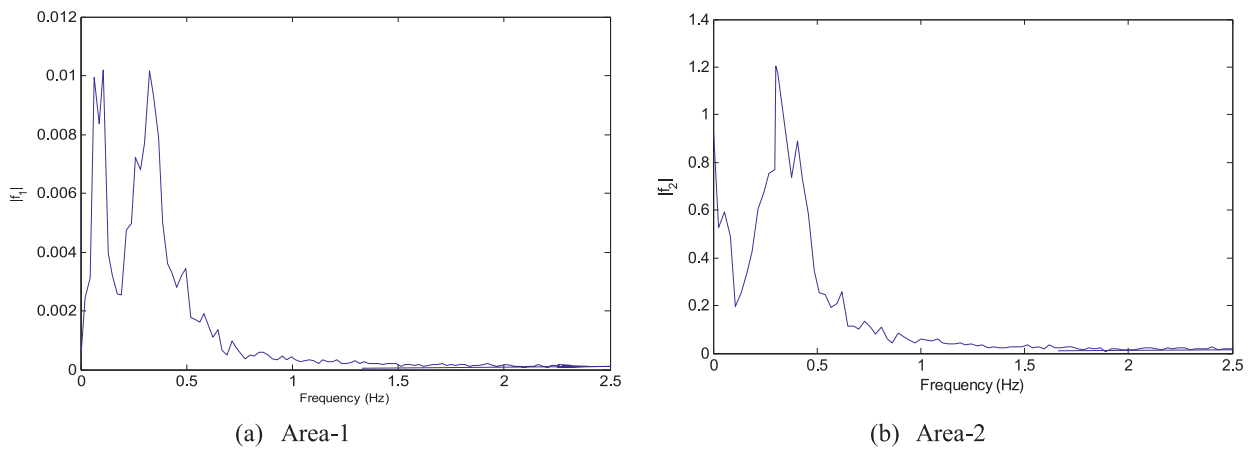


Fig. 14. Frequency spectrum of area-1 & area-2.

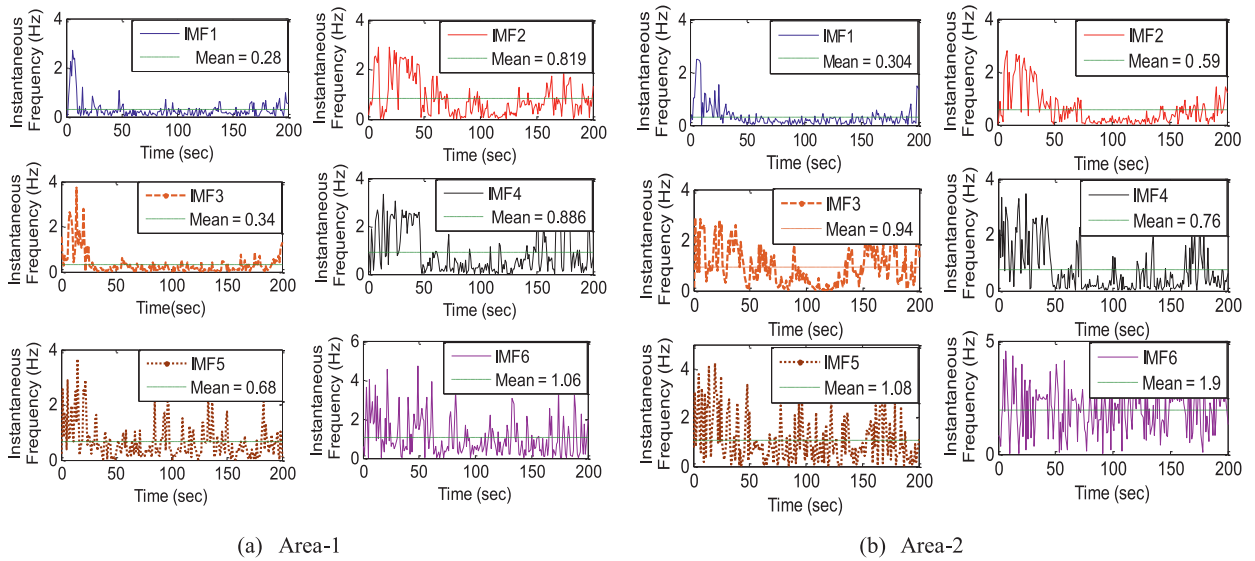


Fig. 15. Instantaneous frequency of interest IMF in union model.

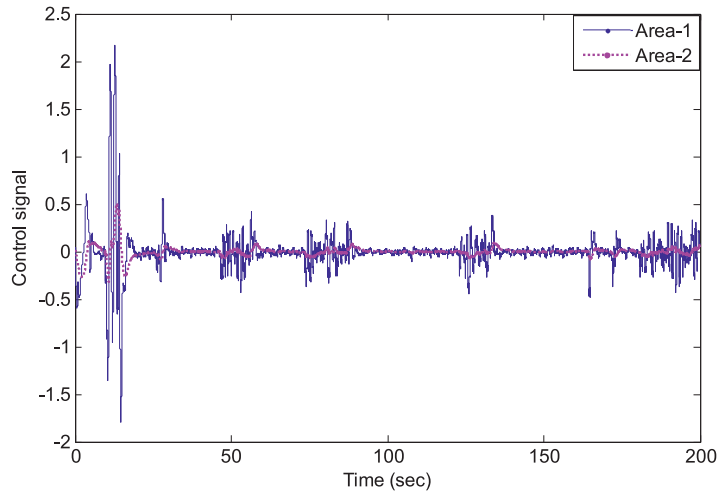


Fig. 16. Area control error signal of area-1 & area-2 two-mass model with thermal system.

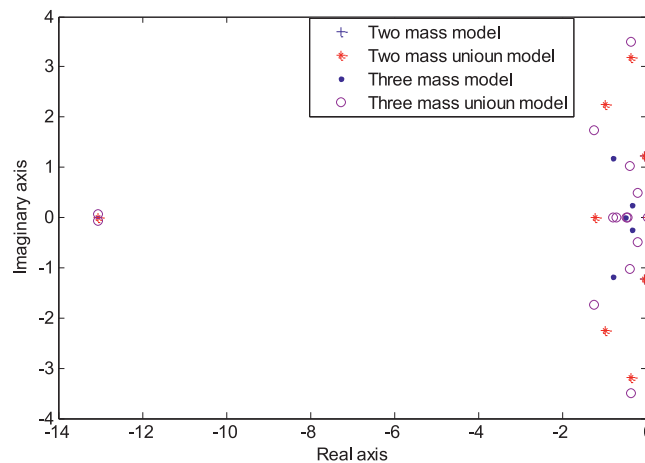


Fig. 17. Location of poles of two-mass/three-mass for models and its union model.

Table 7
Excited Modes.

Signal	Wind Profile-1 (Vander Hoven Model)		Wind Profile –2 Real time test signal	
	Two-mass model	Three-mass model	Two-mass model	Three-mass model
T_{shaft}	0.322 , 0.448, 0.82, 0.79 , 0.88 , and 1.25	0.34, 0.599 , 0.98, 0.76, 0 0.853 , and 1.08	0.4657 , 0.4044, 1.264, 2.035, 2.16, 2.227 , 2.38	0.401 , 0.471, 1.509, 2.09, 2.15, 2.41 , 2.12
Δf_1	0.285 , 0.819 , 0.3418, 0.88 , 0.54, and 1.06	0.467, 0.996, 0.7241, 0.5784 , 0.8771 , 1.617	0.479 , 1.313, 1.98, 2.09, 2.38, 2.27	0.464 , 0.9608, 1.97, 2.17 , 2.38 , 2.214
Δf_2	0.30 , 0.59, 0.94 , 0.76 , 1.08, and 1.36	0.276, 0.39, 1.7, 0.82 , 1.2, 1.60	0.441 , 1.98, 2.23 , 2.22, 2.47	0.427 , 2.05, 2.227 , 2.45 , 2.338 , 2.35
ΔP_{tie}	0.34 , 0.85 , 1.79, 1.73, 1.948, 2.30, 2.323	0.5281 , 0.845 , 1.76, 1.725, 2.114, 2.32, 2.26, 2.47	0.468 , 2.34, 2.15, 2.224 , 2.445	0.4409 , 2.05, 2.196 , 2.45

equals to one decrement in number of masses. So, from the above studies based on two-mass and three-mass wind drive train models, it is concluded that lower oscillation frequency is present in three mass wind turbine model. So, three-mass drive train model is more accurate and results in better transient representation but requires more computational effort. Two-mass model is derived out of three-mass model, which can also significantly represent the dynamics

7. Conclusions

In this study small-signal stability analysis for WT system based on different drive train models (two-mass/ three-mass model) and its union model with integration of thermal power system is investigated. The detail state space modelling is performed for various case studies. The stability study at initial stage was performed using eigenvalue analysis. Further to know the exact relation between modes and corresponding state variable, participation factor was computed. The understanding on excitation of flexible modes, the system is excited by varying nature of wind profile. The FFT is applied to compute the frequency spectrum of signal to confirm the presence of oscillation modes. The decomposition into low frequency oscillation (LFO) modes as a collection of IMFs computed from HHT is also obtained. Subsequently, instantaneous frequency is estimated on IMFs. Two different wind profiles are taken in this study; first one modeled via Vander-Hoven approach and second one as real-time test signal. It was found that frequency mode in union system was approximately close to torsional mode of two-mass and three-mass of drive train WT models. In our future work, load frequency controller design to smoothen the frequency profile in an integrated power system will be carried out.

Conflict of interest

None.

Appendix

Table 8.

Table 8
Parameters of two area thermal power system [25].

	T_t	T_g	B	R	K_p	T_p	T_0
Area-1	0.3	0.1	21	0.05	1	10	0.1986
Area-2	0.4	0.1	21.5	0.05	0.66	7.92	

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